

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Analytical Problem Solving

Code of Conduct:

I V.Madhulika (192312620) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V.Madhulika

Signature of the student

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is -0.5 . For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

Answer:

Net input, $net = (w_1 \times x_1) + (w_2 \times x_2) + \text{bias}$

$net = (0.6 \times 2) + (0.4 \times 1) + (-0.5) = 1.2 + 0.4 - 0.5 = 1.1$

Step activation function: $f(net) = 1$ if $net \geq 0$, and $f(net) = 0$ if $net < 0$.

Since $net = 1.1 \geq 0$, the output $f(net) = 1$.

Conclusion: The student is classified as PASS, since the perceptron output is 1.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Answer:

Error, $e = \text{Target output} - \text{Actual output} = 1 - 0.6 = 0.4$

Delta rule weight update: $\Delta w = \eta \times e \times x$, where η is the learning rate and x is the input.

$\Delta w = 0.1 \times 0.4 \times 2 = 0.08$

New weight, $w(\text{new}) = w(\text{old}) + \Delta w = 0.5 + 0.08 = 0.58$

Conclusion: The weight is updated from 0.5 to 0.58 after one iteration, moving the neuron's output closer to the target value of 1.

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Answer:

TP = 180, TN = 150, FP = 20, FN = 50, Total = 400

Metric	Formula	Calculation	Result
Accuracy	$(TP+TN)/\text{Total}$	$(180+150)/400$	82.5%
Precision	$TP/(TP+FP)$	$180/200$	90.0%
Sensitivity (Recall)	$TP/(TP+FN)$	$180/230$	78.26%
Specificity	$TN/(TN+FP)$	$150/170$	88.24%

Discussion: The model shows high accuracy (82.5%) and strong precision (90%), meaning most predicted positives are correct. However, sensitivity (recall) is only 78.26%, meaning about 22% of actual disease cases (50 false negatives) are missed. In medical diagnosis, false negatives are extremely costly since a missed disease case can delay treatment and endanger the patient. Therefore, although the

model performs reasonably well overall, it is not fully reliable for critical medical diagnosis unless recall is improved further, even if this slightly reduces precision.

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Answer:

Individual (x)	Fitness $f(x)=x^2$
2	4
4	16
6	36
8	64

Selection: Based on fitness, the individuals 8 (fitness 64) and 6 (fitness 36) have the highest fitness scores and are selected as parents.

Crossover: Representing the parents in 4-bit binary — $8 = 1000$ and $6 = 0110$. Exchanging the two least significant bits (LSBs) between the parents gives:

Offspring 1 = 10(from 8) + 10(from 6) = 1010 = 10

Offspring 2 = 01(from 6) + 00(from 8) = 0100 = 4

The crossover thus produces two new candidate solutions (10 and 4) that were not present in the original population.

Role of Mutation:

- Mutation randomly flips one or more bits in an offspring's chromosome, introducing genetic variation that crossover alone cannot produce.
- It maintains diversity in the population, preventing all individuals from converging to the same solution too early (premature convergence).
- It allows the algorithm to explore new regions of the search space and escape local optima, increasing the chance of finding the true global maximum of $f(x) = x^2$.

5. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

Answer:

Analysis: Training accuracy rises continuously to 95%, while validation accuracy peaks around 72% and then falls to 68%. This divergence — training performance improving while validation performance degrades — is the classic signature of overfitting.

Why it occurs: As training continues, the network has enough capacity (parameters) to start memorizing the specific patterns and noise in the training data rather than learning generalizable features. This makes it perform extremely well on data it has already seen but poorly on new, unseen data.

Two techniques to address overfitting:

- Early Stopping – stop training at the epoch where validation accuracy is highest (around 72%), before it starts to decline.
- Regularization / Dropout – add L1/L2 penalties on weights or randomly deactivate neurons during training to prevent the network from over-relying on specific units or large weights.

Significance of validation accuracy: Validation accuracy estimates how well the model generalizes to data outside the training set. It is used to tune hyperparameters, decide when to stop training, and select the best-performing model configuration, making it a crucial indicator of real-world model performance rather than mere memorization.

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is -0.6 . For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected.

Answer:

$$\text{net} = (0.7 \times 1.5) + (0.5 \times 2) + (-0.6) = 1.05 + 1.0 - 0.6 = 1.45$$

Since $\text{net} = 1.45 \geq 0$, applying the step function gives output $f(\text{net}) = 1$.

Conclusion: The loan is **APPROVED**, since the perceptron output is 1.

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration.

Answer:

$$\text{Error, } e = \text{Target} - \text{Actual} = 0.9 - 0.5 = 0.4$$

$$\Delta w = \eta \times e \times x = 0.2 \times 0.4 \times 3 = 0.24$$

$$\text{New weight} = 0.4 + 0.24 = 0.64$$

Conclusion: The updated weight after one iteration is 0.64.

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical.

Answer:

$$\text{TP} = 120, \text{TN} = 200, \text{FP} = 30, \text{FN} = 50, \text{Total} = 400$$

Metric	Formula	Calculation	Result
Accuracy	$(\text{TP} + \text{TN}) / \text{Total}$	$(120 + 200) / 400$	80.0%
Precision	$\text{TP} / (\text{TP} + \text{FP})$	$120 / 150$	80.0%
Recall (Sensitivity)	$\text{TP} / (\text{TP} + \text{FN})$	$120 / 170$	70.59%
Specificity	$\text{TN} / (\text{TN} + \text{FP})$	$200 / 230$	86.96%
F1-Score	$2 \cdot (\text{P} \cdot \text{R}) / (\text{P} + \text{R})$	$2(0.80 \times 0.7059) / (0.80 + 0.7059)$	75.0%

Evaluation: With recall of only 70.59%, the model misses close to 30% of actual positive cases (50 false negatives). In applications where false negatives are critical — such as disease detection or fraud that must not go unnoticed — this level of recall is a significant weakness. Although accuracy and precision are reasonably good, the model would need threshold tuning, class re-weighting, or additional training data to raise recall before it can be considered suitable for such critical use cases.

9. A Genetic Algorithm is used to maximize the function $f(x) = 2x + 3$ for an initial population of values $\{1, 3, 5, 7\}$. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence.

Answer:

Individual (x)	Fitness $f(x)=2x+3$
1	5
3	9
5	13
7	17

Top two candidates selected (highest fitness): $x = 7$ (fitness 17) and $x = 5$ (fitness 13).

Encoding as 3-bit binary: $7 = 111$, $5 = 101$. Performing single-point crossover after the first bit:

Parent 1 = 1|11, Parent 2 = 1|01 $\rightarrow$ Offspring 1 = 1+01 = 101 = 5, Offspring 2 = 1+11 = 111 = 7.

This produces new chromosome combinations by exchanging the genetic material of the two fittest parents.

Role of Mutation in avoiding premature convergence:

Mutation randomly flips a bit in an offspring (e.g., changing 101 to 100), introducing values that neither parent possessed. Without mutation, repeated crossover among a small set of high-fitness individuals can cause the population to converge quickly to a local optimum. Mutation reintroduces diversity, allowing the search to continue exploring the solution space and reducing the risk of the algorithm settling prematurely on a sub-optimal value of x .

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance.

Answer:

Issue: This is a clear case of overfitting. The gap that opens up between the continuously falling training loss and the rising validation loss shows the network is fitting the training data (including its noise) rather than learning generalizable patterns.

Reason: As training progresses beyond the optimal point, the model has enough capacity to memorize specific training examples. This drives training loss down further, but the memorized patterns do not transfer to unseen validation data, so validation loss rises.

Techniques to improve generalization:

- Early Stopping – halt training at the epoch where validation loss is at its minimum.

- Regularization (L1/L2) or Dropout – penalize large weights or randomly deactivate neurons during training so the model cannot over-rely on any single pattern, encouraging a simpler, more generalizable model.

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is -0.4 . For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Answer:

$$\text{net} = (0.5 \times 2) + (0.3 \times 1) + (-0.4) = 1.0 + 0.3 - 0.4 = 0.9$$

Since $\text{net} = 0.9 \geq 0$, the step function gives output $f(\text{net}) = 1$.

Conclusion: The customer IS predicted to make a purchase, since the perceptron output is 1.

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration.

Answer:

$$\text{Error, } e = \text{Target} - \text{Actual} = 1 - 0.7 = 0.3$$

$$\Delta w = \eta \times e \times x = 0.15 \times 0.3 \times 2.5 = 0.1125$$

$$\text{New weight} = 0.6 + 0.1125 = 0.7125$$

Conclusion: The weight is updated from 0.6 to 0.7125 after one iteration of training.

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly.

Answer:

$$\text{TP} = 140, \text{TN} = 180, \text{FP} = 25, \text{FN} = 35, \text{Total} = 380$$

Metric	Formula	Calculation	Result
Accuracy	$(\text{TP} + \text{TN}) / \text{Total}$	$(140 + 180) / 380$	84.21%
Precision	$\text{TP} / (\text{TP} + \text{FP})$	$140 / 165$	84.85%
Recall (Sensitivity)	$\text{TP} / (\text{TP} + \text{FN})$	$140 / 175$	80.0%
Specificity	$\text{TN} / (\text{TN} + \text{FP})$	$180 / 205$	87.80%
F1-Score	$2 \cdot (\text{P} \cdot \text{R}) / (\text{P} + \text{R})$	$2(0.8485 \times 0.80) / (0.8485 + 0.80)$	82.35%

Analysis: For fraud detection, false positives (25 cases, legitimate transactions flagged as fraud) create customer friction and operational cost, so precision matters greatly. At 84.85%, precision is reasonably good but not perfect — about 15% of flagged cases are false alarms. The model is moderately suitable for this application, but threshold tuning or additional features could further reduce false positives and improve precision without unduly sacrificing recall.

14. A Genetic Algorithm is applied to maximize the function $f(x) = 3x + 2$ for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a

crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.

Answer:

Individual (x)	Fitness $f(x)=3x+2$
2	8
5	17
7	23
9	29

Best candidates selected: $x = 9$ (fitness 29) and $x = 7$ (fitness 23).

Encoding as 4-bit binary: $9 = 1001$, $7 = 0111$. Single-point crossover after the 2nd bit:

Parent 1 = $10|01$, Parent 2 = $01|11 \rightarrow$ Offspring 1 = $10+11 = 1011 = 11$, Offspring 2 = $01+01 = 0101 = 5$.

The crossover produces two new candidates (11 and 5) extending the search beyond the original population.

Role of Mutation:

Mutation randomly flips a bit in an offspring's chromosome (e.g., $1011 \rightarrow 1111 = 15$), generating values that could not arise from crossover of the current parents alone. This keeps the population genetically diverse, prevents the algorithm from converging too early around a narrow set of values, and gives it the ability to escape local optima in the search for the true maximum of $f(x) = 3x + 2$.

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation.

Answer:

Issue: The widening gap between training accuracy (climbing to 97%) and validation accuracy (plateauing/fluctuating around 72%) indicates overfitting.

Why it happens: With enough capacity and training time, the model begins to memorize fine-grained, training-specific patterns and noise instead of the broader, generalizable features needed for unseen data, so gains on the training set no longer translate into gains on the validation set.

Suggested techniques:

- Dropout – randomly deactivates a fraction of neurons in each training pass, preventing the network from co-adapting to specific units and forcing it to learn more robust, distributed representations.
- Data Augmentation – artificially expands the training set (e.g., rotations, flips, noise for images) so the model sees more variation and is less likely to memorize a fixed set of examples.

Importance of validation metrics: Validation accuracy/loss provide an unbiased estimate of how the model will perform on new, real-world data. They guide critical decisions such as when to stop training, which hyperparameters to choose, and which model checkpoint to deploy, making them essential for building models that generalize rather than merely fit the training set.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	25	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand